Integrating LLMs and Digital Twins for Adaptive Multi-Robot Task Allocation in Construction

Min Deng, *Member, IEEE*, Bo Fu, Lingyao Li, and Xi Wang, *Member, IEEE*

***Abstract*—Multi-robot systems are emerging as a promising solution to the growing demand for productivity and safety across industrial sectors. However, effectively coordinating multiple robots in dynamic and uncertain environments, such as construction sites, remains a challenge, particularly due to unpredictable factors like material delays, unexpected site conditions, and weather-induced disruptions. To address these challenges, this study proposes an adaptive task allocation framework that strategically leverages the synergistic potential of Digital Twins, Integer Programming (IP), and Large Language Models (LLMs). The multi-robot task allocation problem is formally modeled using IP, considering task dependencies, robot heterogeneity, scheduling constraints, and replanning needs. A narrative-driven schedule adaptation mechanism is introduced, where unstructured natural language inputs are interpreted by an LLM to autonomously update optimization constraints, enabling human-in-the-loop flexibility without manual programming. A digital twin system enables real-time synchronization between physical operations and their digital representations and serves as the user interface, creating a closed-loop feedback mechanism that keeps the system responsive to site changes. A case study demonstrates both the computational efficiency of the optimization algorithm and the reasoning performance of LLMs, achieving over 97% accuracy in constraint and parameter extraction. The results confirm the practicality, adaptability, and cross-domain applicability of the proposed methods.**

***Note to Practitioners*—Construction projects often face sudden disruptions, such as changing priorities, late material arrivals, or weather delays, that can affect planned robot operations. This work presents a practical approach through a real-time site digital twin, connected to a language-based interface that understands plain human instructions. The system is integrated with a scheduling module that can allocate and replan tasks based on human instructions. This enables site managers to adapt robot assignments instantly without modifying the code manually, preserving operational efficiency and minimizing downtime. While the current simulation setup simplifies certain aspects of real-world construction scenarios, the framework is designed for straightforward integration with existing multi-robot and human-robot collaborative construction systems, making it well-positioned for near-term industry adoption. Beyond construction, it can benefit manufacturing, logistics, and other domains requiring flexible multi-agent coordination.**

***Index Terms*—Construction robotics, multi-robot systems, task allocation, scheduling and coordination, digital twins.**

Received 15 August 2025; revised 27 December 2025 and 17 March 2026; accepted 5 May 2026. Date of publication 28 May 2026; date of current version 4 June 2026. This article was recommended for publication by Associate Editor D. Gurdur Broo and Editor J. Gu upon evaluation of the reviewers' comments. *(Min Deng and Bo Fu contributed equally to this work.) (Corresponding authors: Xi Wang; Lingyao Li.)*

Min Deng is with the Department of Civil and Environmental Engineering, University of Tennessee, Knoxville, TN 37996 USA (e-mail: mindeng@utk.edu).

Bo Fu is with Amazon Robotics, North Reading, MA 01864 USA (e-mail: bofu@amazon.com).

Lingyao Li is with the School of Information, University of South Florida, Tampa, FL 33620 USA (e-mail: lingyaol@usf.edu).

Xi Wang is with the Department of Construction Science, Texas A&M University, College Station, TX 77843 USA (e-mail: xiwang@tamu.edu).

This article has supplementary downloadable material available at https://doi.org/10.1109/TASE.2026.3697962, provided by the authors.

Digital Object Identifier 10.1109/TASE.2026.3697962

I. Introduction

With rising demands for faster project delivery and improved efficiency, automation is becoming an essential solution for the construction industry [1], [2], [3]. Robotics, particularly the use of coordinated teams of robots, offers a promising approach that could revolutionize traditional construction practices. Robotic systems are being employed on construction sites to assist with tasks such as material delivery [4], assembly [5], [6], [7], and installation [8], [9], with the potential to significantly improve efficiency [10], [11] and safety [12]. However, effectively coordinating multiple robots in dynamic and unpredictable construction environments remains a considerable challenge. Ensuring smooth interoperability and avoiding operational conflicts require sophisticated algorithms and management strategies [13], [14], [15]. In addition, the limited number of available robots and their diverse capabilities add complexity to task assignment. This necessitates task allocations that are not only efficient but also aligned with each robot's specialized skills and functional strengths, thereby making the problem NP-hard [16].

Although existing solvers and algorithms based on standard Integer Programming (IP) [17] and Genetic Algorithms (GA) [18] can address task allocation problems in multiple scenarios, they lack the flexibility to adapt to changes in construction sites. When implemented, these methods are generally designed to find solutions under static constraints [19], [20], [21]. Uncertainties in construction environments may arise from various sources, including human decision-making, abrupt weather shifts, delays in material supply, and operational disruptions. These factors can influence task priorities, durations, or dependencies. For example, sudden weather changes, such as an unexpected rainstorm, might force the suspension of outdoor tasks [22], [23]. Operational events like unforeseen robot malfunctions or the deployment of new robots require recalculations to maintain optimal task assignments. However, purely mathematical algorithms lack the ability to reason about these real-world construction site dynamics. Therefore, recalculating task schedules in response to these changes often requires extensive manual intervention and algorithmic reconfiguration, which is time-consuming and impractical. As a result, the existing task allocation approaches might result in project delays [23], underutilized labor [24],

1558-3783 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

[25], and resource inefficiencies [26]. In addition to the lack of reasoning capability, current research also lacks the ability to understand task requirements in real time based on actual construction progress. These limitations and gaps highlight the need for flexible approaches capable of adapting to the dynamic and ever-changing conditions of construction sites, leading to the following research questions:

- **RQ1.** What task allocation approaches can effectively handle dynamic characteristics and uncertainties inherent in multi-robot operations on construction sites?
- **RQ2.** How can real-time task information from construction progress be used to dynamically adjust task assignments for multiple robots?

To address these research questions, this study introduces an optimization framework for multi-robot task allocation (MRTA) that integrates digital twin, large language models (LLMs) [27], and IP. It continuously integrates real-time sensor inputs and robot state information to update task statuses, site conditions, and resource availability, ensuring that optimization constraints remain consistent with current field conditions. At the same time, it interprets unstructured human instructions through an LLM-based reasoning module and translates them into structured constraint updates for the IP model. This unified mechanism enables adaptive, human-in-the-loop task allocation under evolving site conditions without manual reprogramming. The key contributions of this study are as follows:

1) We develop a digital twin-based framework that enables communications between real-time robotic operations and digital building models (e.g., Building Information Models (BIMs)). This integration supports the dynamic updating of task statuses and material supplies for task management and robot control.
2) We formulate the MRTA problem as an IP model tailored to the complexities of dynamic construction tasks. The model accounts for task interdependencies, heterogeneous robot capabilities, and evolving constraints.
3) We propose a novel mechanism for narrative-driven task allocation adaptation, whereby an LLM interprets unstructured natural language inputs and autonomously modifies optimization constraints. This approach facilitates human-in-the-loop decision-making while eliminating the need for manual reconfiguration of the optimization code.
4) We validate the proposed framework through extensive simulation-based experiments. They demonstrate both the computational efficiency of the optimization methods and the reasoning accuracy of multiple LLMs. The results demonstrate the framework's effectiveness, adaptability, and potential for practical deployment in real-world construction scenarios.

The rest of this paper is organized as follows. Section II reviews the related literature and identifies key research gaps. Section III presents the detailed research methodology, including the design of the closed-loop digital twin system, the formulation of the optimization function, and the development of the LLM-driven adaptive decision-making mechanism. Section IV demonstrates and validates the proposed system and algorithms through a case study, with further discussion provided in Section V. Section VI concludes the paper.

II. Related Work

A. Robotic Automation in Construction

Recent studies have explored the potential of integrating robots into construction tasks [28]. Exploratory studies, such as Yu et al. [29], investigated long-horizon robotic drywall installation tasks through LLM-based modularized skill transfer. Külz et al. [30] enabled rapid robot hardware adaptation to meet specific demands of construction tasks and successfully demonstrated the system for autonomous drilling. To further improve the efficiency, precision, and adaptability of robots, many researchers have also started to implement the concept of digital twins [31], [32]. For example, a BIM-integrated system that enabled task planning and simulation for construction robots in indoor wall painting was developed by Kim et al. [33]. They linked BIM data with ROS and facilitated robot behavior simulation based on construction schedules and site conditions. Wang et al. [34] presented a BIM-integrated framework that can dynamically update an interactive digital twin from BIM data, enabling robots to adjust work plans in response to site deviations with human oversight. The system combined human flexibility with robotic precision to improve construction robustness by demonstrating drywall and block placement tests. Moreover, a digital twin-enabled safety monitoring system for worker-robot collaboration on construction sites was proposed in [35]. By integrating workers, robots, and manipulated materials, the digital twin dynamically calculates separation distances for safety monitoring.

While these studies have demonstrated the potential of implementing robots to assist with construction tasks, they primarily focus on single-task or narrowly scoped applications, without considering the scenarios when a variety of tasks and multi-robots exist. Additionally, existing approaches often require manual adjustments to accommodate evolving site conditions, highlighting a gap in automation for adaptive task management.

B. Algorithms for MRTA

Driven by industrial demands and future trends, MRTA has emerged as a significant research area in robotics, focusing on the efficient coordination and assignment of tasks among multiple robots to accomplish complex objectives [36]. Existing studies have explored various optimization methods to enhance performance metrics such as task completion time [37], [38], [39] and resource utilization [40], [41]. For example, game theory was often used for solving the MRTA, such as the study conducted by Martin et al. [42], which employed the Shapley value to measure the individual contributions of robots and tasks. The authors partitioned complex MRTA problems into smaller subproblems by clustering robots and tasks based on their Shapley values, effectively balancing robot capabilities and task demands. The simulations demonstrated that this framework outperformed conventional metaheuristic algorithms, such as Genetic Algorithms (GA), in both solution

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

quality and computational efficiency. In addition, Yan and Di [43] introduced a hyper-heuristic algorithm for MRTA problems involving compulsory and optional ("functional") tasks. The developed algorithm assigned functional tasks using an influence diffusion model optimized by Particle Swarm Optimization (PSO). Simulation results demonstrated superior performance compared to existing benchmarks, particularly with increasing numbers of functional tasks.

With more complex scenarios, a stochastic programming framework was proposed by Fu et al. [13] to simultaneously optimize task decomposition, assignment, and scheduling for heterogeneous robot teams. Unlike conventional approaches that assumed fixed task decompositions, this framework captured uncertainties in task requirements and agent capabilities using stochastic representations. It enhanced robustness and flexibility by minimizing risk through Conditional Value at Risk (CVaR). The results demonstrated scalability, low optimality gaps, and effective trade-offs among energy, time, and success probabilities. Furthermore, some recent works have investigated static, logistics-oriented task assignment problems using heuristic or genetic-algorithm-based strategies, such as truck–drone coordination and multi-UAV delivery scheduling. In these settings, route planning and load constraints are optimized through decoupled or hybrid multi-population GA formulations [44], [45]. Other studies have employed multi-agent deep reinforcement learning to optimize UAV coordination under communication, energy, and charging constraints in 6G-enabled network environments [46]. These approaches enhance UAV trajectory planning and task completion performance under uncertainty [47].

Despite considerable advancements, existing MRTA approaches often fail to adequately address dynamic uncertainties and unforeseen events prevalent in real-world scenarios, such as construction environments. Such uncertainties often necessitate human-like reasoning to manage effectively, as they can alter task dependencies, initiation times, and task durations.

C. Applications of LLMs in Construction

The integration of LLMs into the construction industry has emerged as a transformative area of research, reshaping traditional approaches to data management, decision-making, and communication. Recent studies have explored the potential of LLMs to enhance construction information exchange [48], planning [49], and inspection [50] through their advanced information processing capabilities. For example, Chen et al. [51] introduced a Meet2Mitigate (M2M) framework to automate the capture, transcription, and analysis of construction meetings by integrating speaker diarization, automatic speech recognition (ASR), and LLMs with retrieval-based summarization. The M2M prototype showed significant accuracy, promising a seamless, automated solution for meeting recaps that enhances project management and decision alignment across teams. This study confirmed LLMs' ability to extract essential information from fragmented content and demonstrated their potential to obtain real-time construction site information through communicating with humans. In addition, the capabilities of LLMs in information integration were investigated by Forth and Borrmann [52], who proposed a methodology to automate the enrichment of BIMs with missing data. By leveraging Semantic Textual Similarity (STS) and fine-tuned LLMs, the approach enabled efficient mapping of thermal properties to construction elements, such as walls and spaces, based on architectural and mechanical, electrical, and plumbing (MEP) data. Tested on real-world case studies, the framework demonstrated high accuracy in construction-specific matching tasks. Moreover, Prieto et al. [53] explored the potential of LLM for generating construction schedules, finding that it can produce coherent and logically organized schedules that fulfill project requirements. Participants reported a positive interaction experience with the tool. One step further, Kim et al. [54] developed a framework integrating LLMs, BIM, and robotic systems to enhance autonomous task planning in construction, aiming to address challenges with dynamic site conditions. The prototype achieved high accuracy in managing task responses and adapting to environmental changes, demonstrating the potential for improving task adaptability and efficiency in complex construction environments. In terms of construction site inspection, a framework named AutoRepo, which could use multimodal LLMs and autonomous vehicles to automate report generation, was presented by Du et al. [50]. It aimed at improving efficiency and reducing human error in traditional inspection practices. AutoRepo had shown the potential to deliver high-quality, standardized reports with greater accuracy and speed. The reports generated by the framework were deemed satisfactory based on expert reviews.

Although existing studies demonstrate promising applications of LLMs in construction-related tasks, they lack a comprehensive framework for dynamic multi-robot task management in rapidly changing construction environments. Specifically, they did not leverage the potential of LLMs to autonomously reassign and adapt tasks in response to real-time environmental factors. Additionally, while some frameworks incorporate structured data sources such as BIMs, these models primarily serve as static information databases for data retrieval, annotation, or schedule generation. These systems are typically limited to offline or single-platform operations and lack real-time interaction with physical robots or dynamic site feedback.

III. Methodology

The methodology developed in this study is illustrated in Fig. 1. It introduces a scalable framework for situation-aware decision-making in multi-robot task allocation, enabled by an LLM. The system is capable of operating with or without an initial digital model (e.g., BIM) and has a closed-loop feedback mechanism that synchronizes the physical site with its digital representation.

The digital twin serves as a bridge for synchronization between the physical site and the digital model. The digital model integrates: (1) a list of construction tasks and their precedence relationships that can either be presented in the form of a structured list or be automatically extracted from the BIM; (2) site information (i.e., material supplies and locations), which can either be manually entered or automatically

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

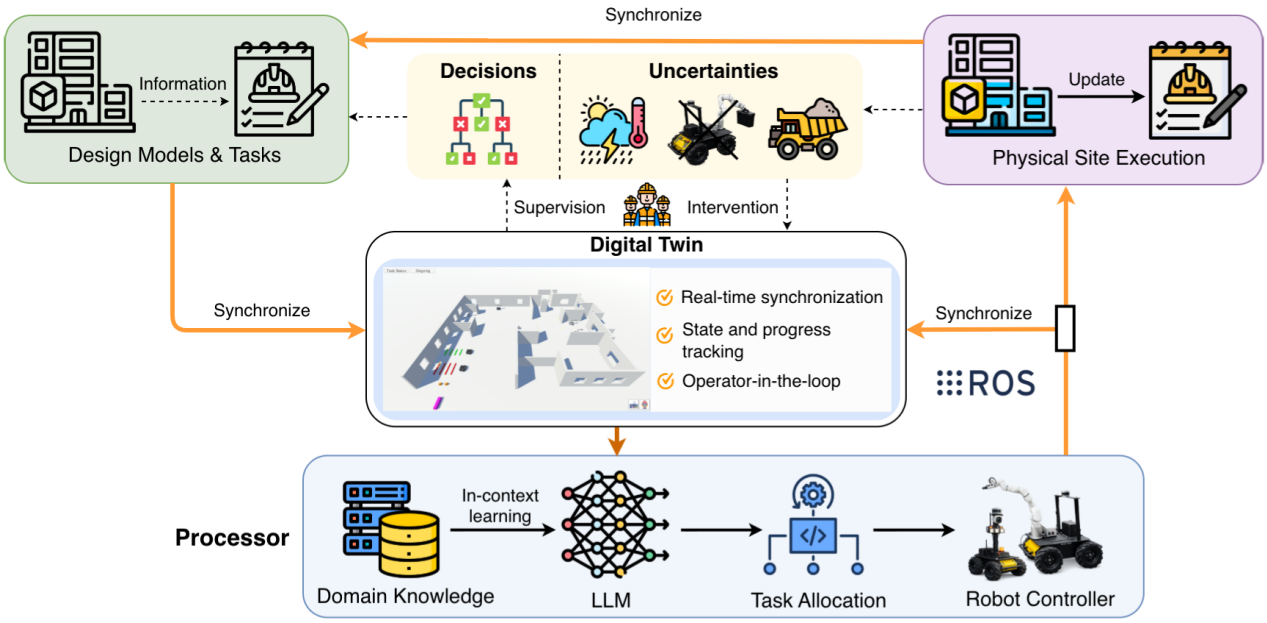


Fig. 1. LLM and digital twin enhanced dynamic robot task allocation.

detected by robots on the physical site and then transmitted to the digital model; and (3) building information that includes object geometries, types, and poses. It includes the completed structures as well as placeholders for components that have not yet been constructed. Each component is associated with attribute information, including its name (reference ID), layer (unbuilt, as-built, materials, or inactive), material type, installation pose, and gripping pose. Based on this information, a comprehensive task list that contains the completion status can be extracted. It covers all active tasks (i.e., tasks that are scheduled to be performed in this project stage), including both the ones that are completed and the pending ones waiting to be assigned and executed.

Human supervision plays a critical role in interpreting construction progress and intervening when unforeseen uncertainties occur, such as weather disruptions, equipment failures, or site access limitations. When human workers identify that changes are needed, they initiate updates through the digital twin interface by providing a natural-language narrative describing the required adjustments. Once the digital twin receives the narrative, it forwards the text to the LLM module in the central processor for interpretation using ROS messages routed through Rosbridge.

The central processor coordinates the intelligent decision-making layer. Here, an LLM interprets contextual inputs (i.e., natural language updates from human operators) with support from a domain knowledge repository. Rather than solving the optimization problem directly, the LLM identifies the relevant parameters. It then translates the narrative into structured constraint updates for the IP model in the task allocation module. The IP model then performs replanning using the updated constraint sets to generate an optimized task allocation plan. The updated task allocation plan is then sent to the robot controllers, which execute the tasks accordingly. Completion data flows back to the digital twin and task model, ensuring continuous updates and alignment between the virtual and physical environments.

A. Digital Twin System

As a critical component of the proposed framework, the digital twin system integrates the digital model and the system central processor to continuously monitor and manage the project's status. It also functions as the user interface, connecting human users with system backends. This enables users to visualize, supervise, and intervene in the project using natural language via keyboard or voice input to submit narrative updates that modify task allocation constraints. The digital twin state at time t is formalized as:

$$S_t = D_0 + \mathcal{F}(S_{t-1}, R_t, SI_t, \Delta_{task}, \Delta_{robot}) \tag{1}$$

Here, S_t represents the current state of the digital twin at time t, D_0 denotes the initial information from the digital model D, $\mathcal{F}$ is the state update function, R_t and SI_t denote the robot states (i.e., arm joint states and base locations and orientations) and site information at time t, respectively, and Δ_{task} and Δ_{robot} capture updates to task statuses and robot statuses (i.e., high-level task each robot in the team is performing) at time t, respectively.

Fig. 2 illustrates the system architecture and information flow of the technical implementation. The digital twin integrates four modules, including the synchronized visualization module, task status tracker, robot status tracker, and user command receiver. These modules enable the digital twin to support three core system functions: (1) *Visualization* provides a unified, real-time representation of the task environment and robot states, allowing human users to observe site conditions as they evolve; (2) *Supervision* enables users to monitor task progress and robot assignments through updated task and robot status streams; and (3) *Intervention* allows users to modify task allocation constraints through natural-language inputs, which

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

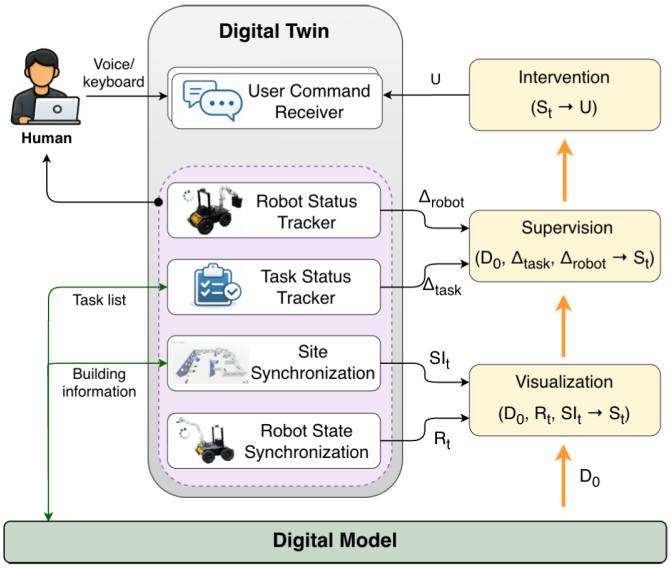


Fig. 2. Digital twin system framework.

are forwarded from the digital twin to the LLM module for automated interpretation and decision-making. The technical details for each function are shown below.

1) Visualization ($D_0, R_t, SI_t \rightarrow S_t$): The synchronized visualization module, adapted from [34], is responsible for continuously updating the visual representation of the site and robots in the digital twin based on SI_t and R_t. Before the process starts, robot emulators are generated by transmitting the URDF models of the robots and associated mesh files from ROS to the digital twin using the ROS# library. In the initial state S_0, the basic digital twin scene is generated from D_0, which includes both geometric and attribute information. In this step, component geometries received from D are instantiated at the corresponding locations in the digital twin as the environment. Attribute information embedded in D is transmitted to the digital twin to adjust the properties of the corresponding components, such as its name, layer, attachment offset, and visualization (e.g., color).

During runtime, the robot emulators mirror on-site robots by subscribing to their state data R_t from ROS, thereby reflecting synchronized robot states to human users. Specifically, each virtual robotic arm updates its joint states according to the subscribed ROS messages and adjusts dependent links accordingly, while transforms of the mobile robot bases are similarly subscribed and used to update the virtual robot base poses. SI_t in this study focuses on construction materials and the evolving as-built environment. Mesh models of construction materials are extracted from their BIM components. When materials are detected on site, the digital twin instantiates their meshes at the subscribed poses. Once a robot grasps a material, it is attached to the robot's end-effector and moves accordingly throughout the manipulation process. After installation, the mesh is detached and added to the building model, with its layer updated from "materials" to "as-built". The corresponding uninstalled component is removed from the "unbuilt" layer, ensuring that the digital twin continuously reflects the evolving construction state.

2) Supervision ($D_0, \Delta_{Task}, \Delta_{Robot} \rightarrow S_t$): The supervision function allows the users to quickly get an oversight of construction progress and multi-robot operations, thereby facilitating intervention and decision making. The capability is implemented through two coordinated modules, robot status tracker and task status tracker. At $t = 0$, the task status tracker extracts from D a comprehensive list of tasks involved. According to the progress, each task is marked as *uninitiated*, *ongoing*, or *completed*. As construction progresses, these labels are updated according to Δ_{task}, which is achieved through communication with the robot controller in ROS. When a robot begins executing a task, it publishes a message on the designated ROS topic, prompting the digital twin to change the corresponding task label from *uninitiated* to *ongoing*. Meanwhile, the robot status tracker updates the operation status of each robot using the task name. Once a completion message is received, the corresponding task status is updated to *completed*, and the robot status remains synchronized with the task it is currently performing. If a robot is not executing any task, its status is marked as *idle*.

3) Intervention ($S_t \rightarrow U$): The user command receiver module supports user interventions when users make intervention decisions U during the visualization and supervision process or uncertainties occur. In these cases, users can submit high-level instructions in natural language by keyboard typing or directly through voice. The narrative text is sent to the LLM through Rosbridge using the ROS# library and processed by the embedded LLM into actionable constraint modifications. These modifications are then forwarded as valid inputs to the task allocation module, which updates robot task assignment.

After the completion of tasks or upon reaching a project milestone, the updated task list and building information are transmitted to update the digital model. This information includes components installed during construction and changes in site conditions, such as leftover construction materials on-site. As a result, the digital model maintains an up-to-date record and accurate representation of the project. This closed-loop feedback mechanism ensures that the digital model accurately reflects the project's as-built condition. It also enhances the model's utility in subsequent phases of work, such as facility management, progress auditing, and future renovations.

B. Multi-Robot Task Allocation: Plan Creation

This section formally defines the task allocation and scheduling problem, formulates it as an integer program, and describes the algorithms used to solve it.

Suppose there are $n_{\mathcal{A}}$ capabilities, denoted by $\mathcal{A} = \{1, \ldots, n_{\mathcal{A}}\}$, $n_{\mathcal{R}}$ heterogeneous robots, denoted by $\mathcal{R} = \{1, \ldots, n_{\mathcal{R}}\}$, and $n_{\mathcal{T}}$ tasks, denoted by $\mathcal{T} = \{1, \ldots, n_{\mathcal{T}}\}$. Each robot has a subset of capabilities, and each task requires a team of robots with the required capabilities to serve it. When a team of qualified robots $\mathcal{R}_i \subset \mathcal{R}$ arrive, the task $i \in \mathcal{T}$ can be completed after time duration T_i^D. A robot may only serve one task at a time. Additionally, each task may have a set of dependencies $\mathcal{T}_i \subset \mathcal{T}$, i.e., other tasks that must be completed beforehand. The goal is to schedule tasks on a

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DEFINITION OF THE NOTATION

	Meaning
x_{ir}	= 1 if task $i \in \mathcal{T}$ is assigned to robot $r \in \mathcal{R}$.
t^s_{ir}	The start time if robot $r \in \mathcal{R}$ works on task $i \in \mathcal{T}$.
t^e_{ir}	The end time if robot $r \in \mathcal{R}$ works on task $i \in \mathcal{T}$.
t^s_i	The start time of task $i \in \mathcal{T}$.
t^e_i	The end time of task $i \in \mathcal{T}$.
y_{ijr}	Auxiliary variables for robot scheduling.
y_{ij}	Auxiliary variables for task scheduling conflict.
z_{ijr}	Auxiliary activation variable for robot scheduling constraints.
T^D_i	The time duration to complete task $i \in \mathcal{T}$.
T^s_i	The earliest start time for task $i \in \mathcal{T}$.
T^e_i	The latest end time for task $i \in \mathcal{T}$.
T_{large}	A large time constant.
a_{kr}	The amount of capability $k \in \mathcal{A}$ available on robot $r \in \mathcal{R}$.
b_{ki}	The amount of capability $k \in \mathcal{A}$ required to execute task $i \in \mathcal{T}$

set of available robots in a way that minimizes the makespan, the time by which all tasks have been completed.

We formulate the above-mentioned task allocation and scheduling problem as an integer program. Here, we provide common notations in Table I. The decision variables are x_{ir} for task assignment and t^s_{ir}, t^e_{ir}, t^s_i, and t^e_i for scheduling.

1) Objective Function: The objective function jointly minimizes the makespan and the individual task completion times. The third piece in the objective function penalizes the number of robots assigned to tasks. The makespan is defined as the maximum end time among all tasks. We set $C_m \gg C_s \approx C_r$ to ensure that the makespan is the primary objective.

$$\min_{\substack{x_{ir},\ t^s_i,\ t^e_i,\\ t^s_{ir},\ t^e_{ir}}} \left(C_m \max_{i\in\mathcal{T}} \ t^e_i + C_s \sum_{i\in\mathcal{T}} \ t^e_i + C_r \sum_{i\in\mathcal{T}} \sum_{r\in\mathcal{R}} x_{ir} \right) \tag{2}$$

2) Variable Bound Constraints: x_{ir} is a binary task allocation variable, equal to 1 if task $i \in \mathcal{T}$ is assigned to a team that contains robot $r \in \mathcal{R}$, and 0 otherwise. The scheduling variables are continuous, with the requirement that the end time is always larger than the start time.

$$x_{ir} \in \{0,1\}, \quad \forall i \in \mathcal{T}, \quad \forall r \in \mathcal{R} \tag{3}$$
$$0 \le t^s_i \le t^e_i, \quad \forall i \in \mathcal{T} \tag{4}$$
$$0 \le t^s_{ir} \le t^e_{ir}, \quad \forall i \in \mathcal{T}, \quad \forall r \in \mathcal{R} \tag{5}$$

3) Task Dependency Constraints: The task dependency is specified using the time variables. If task i depends on the completion of task j, then task i can start only after task j is completed.

$$t^s_i \ge t^e_j, \quad \forall j \in \mathcal{T}_i, \quad \forall i \in \mathcal{T} \tag{6}$$

4) Task Requirement Constraints: A capability model is introduced to define the task requirement constraints. Let $a_{kr} \in \mathbb{R}_{\ge 0}$ denote the amount of capability $k \in \mathcal{A}$ available on robot $r \in \mathcal{R}$, and let $b_{ki} \in \mathbb{R}_{\ge 0}$ denote the amount of capability $k \in \mathcal{A}$ required to execute task $i \in \mathcal{T}$. The robot team assigned to a task must collectively possess all the capabilities required by that task, as enforced by (7).

$$\sum_{r\in\mathcal{R}} a_{kr} x_{ir} \ge b_{ki}, \quad \forall k \in \mathcal{A}, \quad \forall i \in \mathcal{T} \tag{7}$$

5) Task Schedule Constraints: Equation (8) specifies the relation between the task start and end time. The constraints (9)- (12) are organized into two groups. The first group, (9)-(10), ensures that when robot $r \in \mathcal{R}$ is in the team assigned to task $i \in \mathcal{T}$, the task start time t^s_i equals the robot-specific start time t^s_{ir}. Similarly, the second group, (11)-(12), guarantees that the task end time t^e_i matches t^e_{ir} under the same assignment. Note that T_{large} is a large time constant.

$$t^e_i = t^s_i + T^D_i, \quad \forall i \in \mathcal{T} \tag{8}$$
$$t^s_i \le t^s_{ir} + T_{\text{large}}(1 - x_{ir}), \quad \forall r \in \mathcal{R}, \quad \forall i \in \mathcal{T} \tag{9}$$
$$t^s_i \ge t^s_{ir} - T_{\text{large}}(1 - x_{ir}), \quad \forall r \in \mathcal{R}, \quad \forall i \in \mathcal{T} \tag{10}$$
$$t^e_i \le t^e_{ir} + T_{\text{large}}(1 - x_{ir}), \quad \forall r \in \mathcal{R}, \quad \forall i \in \mathcal{T} \tag{11}$$
$$t^e_i \ge t^e_{ir} - T_{\text{large}}(1 - x_{ir}), \quad \forall r \in \mathcal{R}, \quad \forall i \in \mathcal{T} \tag{12}$$

6) Robot Schedule (No-Overlap) Constraints: Each robot can perform at most one task at a time, so their task schedules must not overlap, as specified in (13)-(14). The auxiliary variable z_{ijr} equals 0 when robot $r \in \mathcal{R}$ is assigned to both tasks $i, j \in \mathcal{T}$, thereby activating the no-overlap constraints. When $y_{ijr} = 1$, task i is scheduled before task j; otherwise, task j is scheduled before task i.

$$z_{ijr} = 2 - x_{ir} - x_{jr}, \quad \forall i < j \in \mathcal{T}, \quad \forall r \in \mathcal{R}$$
$$t^e_{ir} \le t^s_{jr} + T_{\text{large}}(1 - y_{ijr} + z_{ijr}), \quad \forall i < j \in \mathcal{T}, \quad \forall r \in \mathcal{R} \tag{13}$$
$$t^e_{jr} \le t^s_{ir} + T_{\text{large}}\ (y_{ijr} + z_{ijr}), \quad \forall i < j \in \mathcal{T}, \quad \forall r \in \mathcal{R} \tag{14}$$
$$y_{ijr} \in \{0,1\}, \quad \forall i < j \in \mathcal{T}, \quad \forall r \in \mathcal{R} \tag{15}$$

7) Time Window Constraints (Optional): The time window constraint is an optional condition that requires a task to be completed within a specified time interval.

$$T^s_i \le t^s_i \le t^e_i \le T^e_i, \quad \forall i \in \mathcal{T} \text{ with a time constraint} \tag{16}$$

8) Task Conflict Constraints (Optional): The task conflict constraints are optional and ensure that two tasks i and j with conflicting resource requirements are not executed concurrently. y_{ij} is a binary auxiliary variable.

$$t^e_i \le t^s_j + T_{\text{large}}(1 - y_{ij}), \quad \forall i < j \in \mathcal{T} \text{ with conflicts} \tag{17}$$
$$t^e_j \le t^s_i + T_{\text{large}}\ y_{ij}, \quad \forall i < j \in \mathcal{T} \text{ with conflicts} \tag{18}$$
$$y_{ij} \in \{0,1\}, \quad \forall i < j \in \mathcal{T} \text{ with conflicts} \tag{19}$$

The above integer program can be solved using a CP-SAT solver. A CP-SAT solver is an optimization engine that combines constraint programming (CP) techniques with Boolean satisfiability (SAT) solving methods. It is designed to efficiently tackle combinatorial optimization problems - such as scheduling, planning, and assignment tasks - by systematically exploring potential solutions while adhering to a set of constraints. In our work, we employ the CP-SAT solver provided by Google OR-Tools to efficiently solve the integer program.

C. LLM-Driven Adaptive Decision-Making

To enable adaptive task reallocation in response to dynamic site conditions, we introduce a formal mapping that captures

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

TABLE II
TYPES OF CONSTRAINT AND PARAMETER CHANGES

Constraint Type	Definition	Label	Parameter
Task Dependency	Adjustments in the sequence or prerequisite relationships among tasks	1	`[task_id, successors, +/-]`
Task Duration	Variations in the estimated time to complete tasks	2	`[task_id, new_duration]`
Task Starting Time	Changes to tasks' earliest or planned start times	3	`[task_id, start_time_change]`
Number of Robot	Variations in robot availability	4	`[new_robot_type_id, robot_number_change]`
Task Conflict Constraints	Tasks that cannot be performed at the same time	5	`[task_id1, task_id2]`

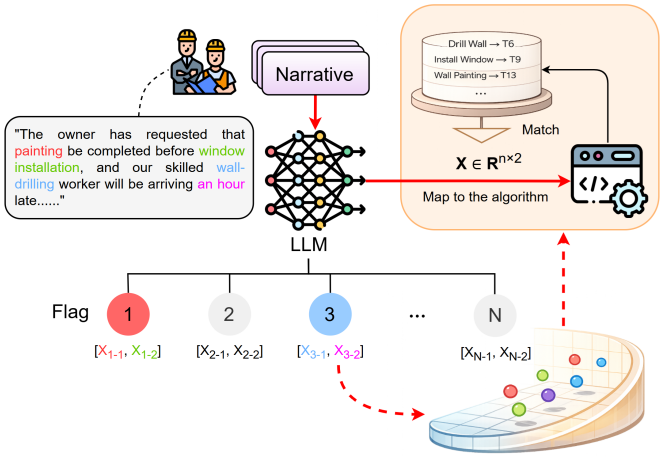


Fig. 3. Pipeline for narrative-driven adaptive schedule optimization.

how the LLM interprets natural language narratives and transforms them into actionable modifications to the task allocation problem. This process serves as the core of the narrative-driven adaptation mechanism, facilitating human-in-the-loop flexibility without requiring direct manipulation of optimization code.

Fig. 3 illustrates the pipeline for enabling adaptive multi-robot task allocation driven by natural language inputs. The process begins with a narrative containing dynamic project updates, such as task sequencing preferences, resource delays, or timing adjustments. An LLM processes this narrative to extract actionable information, identifying relevant task entities (e.g., painting, window installation, wall-drilling) and interpreting relationships such as precedence constraints or temporal shifts. To ensure accurate task reference, a structured database is used to map textual descriptions to corresponding task IDs (e.g., wall-drilling → T6).

Next, the extracted instructions are categorized into discrete flag types, each representing a specific kind of modification as demonstrated in Table II. Each flag is paired with a structured parameter representation, which encapsulates the necessary information to modify the optimization code.

Let the input to the LLM be a narrative description $\mathcal{N}$, such as a sentence or paragraph provided by a site supervisor or worker, describing changes in site conditions, task statuses, or scheduling preferences. The LLM acts as a function:

$$\mathcal{M} : \mathcal{N} \rightarrow \{(C_k, \theta_k)\}_{k=1}^{K} \tag{20}$$

where:

$\mathcal{M}$ is the mapping function learned or encoded by the LLM
C_k is the type of constraint (e.g., dependency, time shift)
θ_k is the parameter set for the update

Each (C_k, θ_k) pair defines a flagged constraint to be dynamically injected into the optimization model. These flags are associated with predefined templates in the integer programming codebase, allowing seamless integration with the solver without altering the base model structure. For example:

- A narrative like "Our skilled wall-drilling worker will be arriving an hour late" will yield $(C = 3, \theta = \{T6, +1\})$.
- A command like "The owner has requested that painting be completed before window installation" maps to $(C = 1, \theta = \{T13 \prec T9\})$.

where T_i represents a task identifier defined in the example case study shown in Table III.

To ensure consistency, the LLM references a structured task knowledge base $\mathcal{T} = \{(T_i, \text{Desc}_i)\}$ that maps textual task names to task IDs. This helps avoid ambiguity and ensures each extracted constraint can be accurately linked to the model.

The LLM output is then parsed into a structured JSON format and passed to the optimization backend, which updates the constraint set accordingly:

$$\mathcal{C}_{\text{new}} = \mathcal{C}_{\text{original}} \cup \{C_k(\theta_k)\} \tag{21}$$

This interaction between the LLM and the optimizer ensures the system can continuously adapt to evolving conditions while maintaining model transparency and interpretability.

Please note that the optimization algorithm operates independently of the LLM. Using an LLM to solve mathematical optimization problems would be inefficient and unreliable due to its limitations in numerical computation and algorithmic convergence. We leverage the LLM's strengths in natural language understanding and reasoning to interpret unstructured human or contextual inputs and translate them into structured updates for the IP model constraints. The actual optimization is executed by a proven solver that guarantees mathematical correctness and reproducibility. This separation offers advantages in the following aspects:

- **Transparency.** The optimization process remains governed by deterministic mathematical models, while all LLM-generated modifications are explicitly logged.
- **Interpretability.** Each natural-language instruction can be traced to its corresponding change in the constraint set, which avoids closed-box decision pathways.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

- **Scalability.** The modular design allows the LLM or solver to be independently upgraded or replaced, ensuring flexibility for larger-scale or domain-specific deployments.

D. LLM Implementation and Prompt Design

Prompt design is critical for LLMs to perform high-quality information extraction, particularly given the intricate nature of construction scheduling tasks [55], [56]. To create effective prompts tailored for construction project scheduling, we utilize prompt engineering strategies articulated by White et al. [55]. These strategies emphasize the necessity of integrating context descriptions, structured templates, and explicit output formatting into prompts. In our scenario, context descriptions detail construction tasks, their interdependencies, durations, and associated robotic capabilities, and we further employ JSON as the structured output format to ensure clarity.

In addition, we incorporate two techniques into our prompt design. The first technique is chain-of-thought (CoT) prompting, which guides the LLM through step-by-step reasoning [57]. This approach can facilitate accurate task identification and constraint recognition by prompting the LLM to engage in deliberate, intermediate reasoning stages before concluding. Specifically, our CoT prompts instruct the LLM to: (1) carefully read the task description; (2) identify the specific task and associated robotic system; (3) determine the constraint type; (4) extract relevant parameters; and (5) format the extracted information. The second technique is few-shot learning. Given that prior research indicates LLMs excel in few-shot learning scenarios [58], [59], we provide multiple examples within our prompt. These examples serve as explicit references, allowing the LLM to capture task requirements more effectively and thereby enhancing the accuracy of its outputs. By leveraging these prompt engineering techniques, we develop a structured prompt template tailored for prompting LLMs to analyze task descriptions of construction project scheduling (see Supplementary Material - Prompt Design.pdf).

For model selection, we refer to the widely recognized Multi-task Language Understanding (MMLU) benchmark [60]. We utilize two popular LLM families, OpenAI's GPT series and Anthropic's Claude series, because of their demonstrated capabilities in complex reasoning and strong performance across diverse NLP tasks [61], [62]. Specifically, from OpenAI, we select GPT-4o-mini, GPT-4o, GPT-4.1-mini and GPT-4.1. From Anthropic, we include Claude-Haiku and Claude-Sonnet.

E. Multi-Robot Task Allocation: Replanning

Suppose the task schedule and robot assignments have already been determined through the task allocation optimization described in the previous section, and the plan has been partially executed. At time T^R, updated information becomes available reflecting changes in task conditions, such as modifications to robot capabilities, availability, expected task durations, additional task dependencies, or time window constraints. These changes may render the current plan infeasible or suboptimal, necessitating the generation of a new plan. At this point, some tasks have been completed, others are in progress, and the rest have not yet started. For the completed and ongoing tasks, their assigned robots and schedules must remain unchanged during the replanning. Furthermore, in many real-world scenarios, modifying the original plan may incur additional operational costs. For instance, altering the schedule may disrupt human worker timelines, resource bookings, and supply deliveries, and rescheduling these elements often requires additional overhead and expenditure. To account for this, penalties can be introduced into the optimization to discourage unnecessary deviations from the original plan.

Given an original task allocation and scheduling plan, represented by the solution to the variables ${}^0x_{ir}$, ${}^0t^s_{ir}$, ${}^0t^e_{ir}$, ${}^0t^s_i$, and ${}^0t^e_i$, let $\mathcal{T}_- = \{i \in \mathcal{T} \mid {}^0t^s_i \le T^R\}$ denote the set of tasks that are ongoing or completed by the replanning time T^R, and let $\mathcal{T}_+ = \{i \in \mathcal{T} \mid {}^0t^s_i > T^R\}$ represent the set of tasks that have not yet started. The replanning optimization is formulated as follows.

1) Replanning Optimization: The replanning objective jointly minimizes the makespan and individual task completion times while also penalizing deviations from the original plan. We set $C_m \gg C_s \approx C_r \approx C_x \approx C_t$ to ensure that the makespan remains the primary objective, with the other terms balanced relative to one another. The constraints include the original constraints (3)-(19), along with additional constraints (25)-(26) to ensure plan consistency.

$$\min_{\substack{x_{ir},\ t^s_i,\ t^e_i,\\ t^s_{ir},\ t^e_{ir}}} \Big(C_m \max_{i\in\mathcal{T}} t^e_i + C_s \sum_{i\in\mathcal{T}} t^e_i + C_r \sum_{i\in\mathcal{T}}\sum_{r\in\mathcal{R}} x_{ir} + C_x\Delta x + C_t\Delta t\Big)$$
$$\text{subject to } (3)-(19) \text{ and } (25)-(26) \tag{22}$$

2) Plan Change Penalty: Changes to the original task assignments and schedules are penalized during replanning to ensure that the new plan accommodates updated task conditions, keeping deviations from the original plan to a minimum. This constraint helps the optimization focus on only those adjustments that are truly necessary to address the disturbance, thereby limiting the scope of modifications and reducing the likelihood of incurring additional operational costs.

$$\Delta x = \sum_{\forall i\in\mathcal{T}_+}\sum_{\forall r\in\mathcal{R}} |x_{ir} - {}^0x_{ir}| \tag{23}$$

$$\Delta t = \sum_{\forall i\in\mathcal{T}_+} \left(|t^s_i - {}^0t^s_i| + |t^e_i - {}^0t^e_i|\right) \tag{24}$$

3) Historical Plan Constraints for Completed Tasks: The robot assignments and task schedules for ongoing or completed tasks must remain unchanged to ensure consistency between the updated plan and the actual execution.

$$t^s_i = {}^0t^s_i, \quad t^e_i = {}^0t^e_i, \quad \forall i \in \mathcal{T}_- \tag{25}$$

$$x_{ir} = {}^0x_{ir} \quad \forall r \in \mathcal{R}, \quad \forall i \in \mathcal{T}_- \tag{26}$$

IV. Case Study

To validate the proposed framework, a case study is conducted to verify the system workflow and evaluate module capabilities. A BIM is employed in this study as the digital environment for evaluating the proposed framework. Rhino

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.



Fig. 4. Simulation of the task allocation and completion.

is used as the BIM platform, which offers an intuitive user interface to edit attribute data and can be interfaced with other applications. The robotic agents receive task assignments based on priority, dependencies, and spatial constraints, ensuring an optimized workflow. Human workers can interact with the system by updating task priorities, modifying scheduling constraints, or assigning specific tasks to robots based on evolving site conditions. As robotic agents complete their tasks, the BIM is updated in real-time, reflecting changes in the construction progress.

Although the construction projects experience a wide spectrum of uncertainties, the relative severity and frequency of these events vary significantly across project types, locations, and operational practices. In this case study, the considered uncertainties, including weather impacts [23], material supply delays [63], equipment malfunctions [64], and task-level disruptions [65], are presented as representative examples rather than an exhaustive set of industry-critical risks. The goal is to demonstrate how the proposed framework adapts to different categories of disruptions by dynamically updating constraint sets in real time. It does not aim to prioritize or quantify specific types of uncertainty. This design choice ensures that the evaluation focuses on the framework's general adaptability and not on the criticality of individual disturbances.

The physical site is simulated in Gazebo,[1] with inter-robot communication established through ROS. Fig. 4 demonstrates an example scenario of the simulation. It is important to note that the robots are not expected to handle all tasks independently, but rather to complete or assist with specific tasks, with human workers often remaining essential to the process. For example, once a robot transfers and places the frame strips on the ground near a window, the associated transfer task is completed, and a human worker installs the frame on the wall. The actions of human workers are not modeled; instead, the world state is updated, and the frames are teleported onto the wall. Following installation, the simulation updates the building state specifications and transmits the relevant information to the digital twin for incorporation into the BIM data.

The multi-platform digital twin integration in the case study is designed to reflect the key components required for practical deployment in real construction environments. BIM provides standardized representations of site geometry, task definitions, and construction sequences that are commonly available in practice. Unity serves as the digital twin and user interface, enabling real-time visualization, monitoring, and human-in-the-loop interaction with the task allocation process. Gazebo simulation is used to emulate robot execution and sensing during validation. In real-world deployment, this component would be replaced by physical robots and on-site sensor streams without altering the higher-level task allocation and decision-making pipeline. Through this integration, the case study validates the system's ability to interpret human narratives, update optimization constraints via the LLM–digital twin integration, and adapt multi-robot task allocation in real time. While Gazebo does not capture all low-level execution complexities of construction robotics, these details are orthogonal to the focus of this work. Robust solutions for robot control, perception, and communication have been extensively studied and can be directly incorporated into the proposed framework.

A. *Digital Twin Demonstration*

The digital twin is developed using Unity, with a project template designed to automatically generate interactive digital twins from different BIMs, eliminating the need for manual setup each time [34]. This template includes five key components: (1) robot emulators for robot state synchronization; (2) BIM object templates for digital model visualization;

[1]Simulation setup: construction materials (e.g., window frames) are assumed to be accurately detectable, and their poses are known exactly by the robots (no detection noise is modeled). Once the gripper contacts a material, the object is assumed to move rigidly with the manipulator and friction is not explicitly modeled. For onboard sensing, the IMU measurements include Gaussian white noise with angular rate standard deviation 2×10^{-4} rad/s (bias mean 7.5×10^{-6} rad/s) and linear acceleration standard deviation 1.7×10^{-2} m/s^2 (bias mean 0.1 m/s^2). The laser scanner measurements include Gaussian white noise with a standard deviation of 0.01 m. The range limits are 0.12-3.5 m with a resolution of 0.015 m, and the scanner provides 360 samples.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

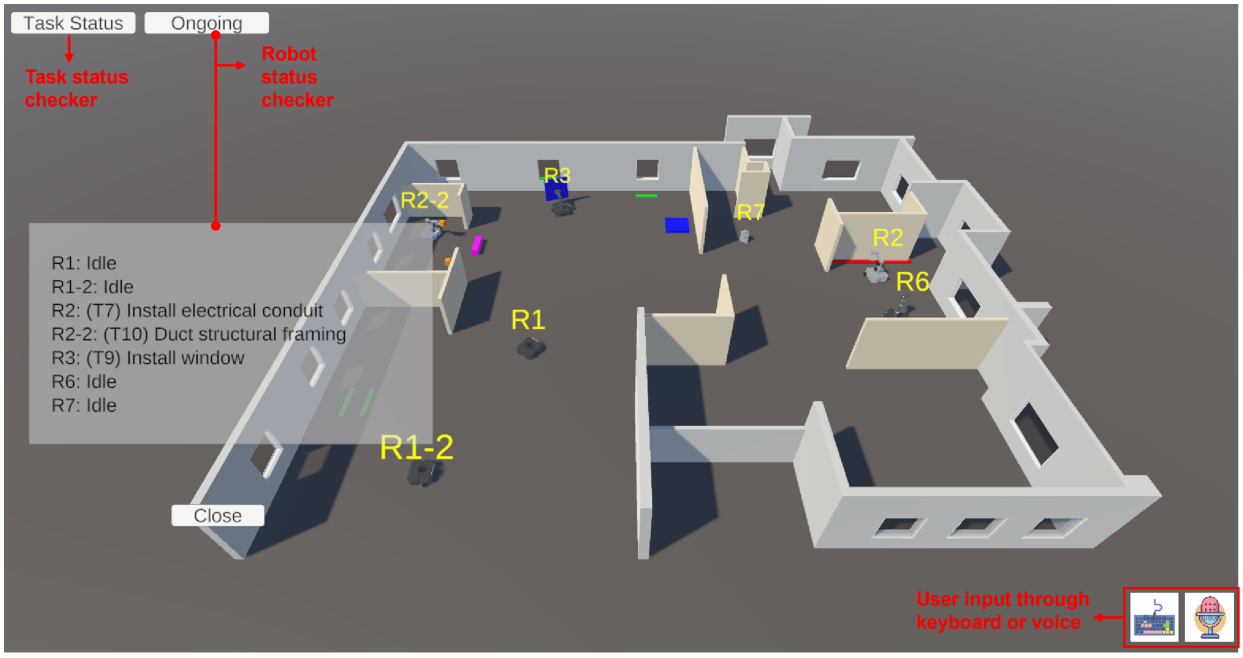


Fig. 5. Screenshot of digital twin interface.

(3) site information templates for site data rendering; (4) user interface elements for user supervision and intervention; and (5) BIM update functions. A screenshot of the digital twin interface and the robot status tracker is shown in Fig. 5.

The robot emulators are created by subscribing to robot descriptions from ROS through the ROS# library [66]. Robot URDF files and meshes, stored in ROS, are retrieved and stored in Unity as game objects. These game objects represent the hierarchical structure of each robot: links are structured as children of their respective parent links, allowing dynamic transformation updates based on real-time state data. A dedicated script is attached to each robot emulator to subscribe to robot states (i.e., arm joint states and base locations and orientations), enabling continuous pose synchronization.

"Prefab" game object templates are used to visualize BIM components. For each component in BIM, a prefab object template is instantiated in the Unity scene. The mesh of the instantiated object is automatically set to the component mesh received from the BIM to replicate the site environment. The script attached to the template can change the name, layer, material, and state of the instantiated object according to its BIM attributes. For example, the objects that have not yet been constructed are rendered as hidden or transparent in the digital twin. This BIM-Unity communication is established with Rhino. Inside, which is an open-source plugin to run Rhino and its API inside other applications on the Windows operating system.

For the site information, this case study specifically focuses on the construction materials. Monitoring the states of materials is essential in construction projects for supply management and task scheduling. Additionally, due to their large size and weight, these materials pose significant safety considerations during robot manipulation. Both the resting and active materials are tracked. Locations of the materials resting on site are detected by robots and instantiated in the digital twin as game objects. For materials being manipulated, they are attached as a child to the end effector or moving platform of the corresponding robot based on their attachment offsets. Therefore, they can move with the robot during operation.

In addition, the user interface elements are created. The digital twin is initiated with two buttons on the top-left corner for users to access the task status tracker and the robot status tracker. Once the "Task Status" button is clicked, a panel shows up with text inside displaying a comprehensive task list along with the status of each task (i.e., uninitiated, ongoing, or completed). The initial task list is received from the BIM at project initiation. As construction progresses, task statuses are updated via messages received from ROS. Similarly, the "Ongoing" button provides the users with a summary of the current task each robot is performing, shown as task IDs, followed by high-level task descriptions, also via messages received from ROS. An example of robot status tracker is demonstrated in Fig. 5. At the bottom-right corner of the UI, two additional buttons are used to collect user inputs. The keyboard button pops up a text input field for typing, while the voice button transcribes spoken input to text with the transcription API from OpenAI. The text collected is then passed to the LLM module for further processing and interpretation.

The human can archive the current project status to the BIM at the project milestone or when the work is finished by pressing a keyboard button. The task statuses are sent to Rhino through COMPAS FAB framework [67] to update the corresponding component attribute information in the BIM. The layer of the components installed during the job period is switched from the "Unbuilt" layer to "As-built" layer, with the model visualization updated according to layer changes. The materials left over on construction sites and their corresponding locations are also sent to be instantiated in the BIM on the "Material" layer for potential future use.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

TABLE III
DURATIONS AND DEPENDENCIES OF THE CONSTRUCTION TASKS

Task	Pred.	Duration	Description	Robot
T_1	-	0.25	Move Electrical Conduit	R_1
T_2	-	0.25	Move Window Frame	R_1
T_3	-	0.25	Move Window	R_1
T_4	-	0.25	Move Duct Structural Materials	R_1
T_5	-	0.25	Move Duct	R_1
T_6	-	0.5	Drill Wall	R_4 / R_2
T_7	T_1, T_6	1	Install Electrical Conduit	R_5 / R_2
T_8	T_2	1	Install Window Frame	R_4 / R_2
T_9	T_3, T_8	0.5	Install Window	R_3
T_{10}	T_4	2	Duct Structural Framing	R_4 / R_2
T_{11}	T_5, T_{10}	2	Install HVAC Duct	R_4 / R_2
T_{12}	T_7	2	Install Wiring	R_5 / R_2
T_{13}	T_{12}	1	Wall Painting	R_6
T_{14}	-	0.5	Construction Site Inspection	R_7

Note: Durations are in hours. "Pred." denotes predecessor.

TABLE IV
ROBOT CAPABILITIES AND QUANTITIES

Robot Number	R_1 [1,4]	R_2 [1,2]	R_3 [1,2]	R_4 [1,2]	R_5 [1,2]	R_6 [1,2]	R_7 [1]
Cargo container	1						
High-payload		1	1	1			
Suction-based gripper			1				
Precise parallel gripper		1			1		
Normal parallel gripper		1		1			
Sprayer						1	
Camera							1
IAQ sensors							1

B. Construction Tasks and Robot Capabilities

Typical construction tasks, such as element delivery, window installation, wall drilling, HVAC duct installation, wiring, wall painting, and site inspections, are used to showcase the capabilities of the methodologies developed in this study. The assumed durations and dependencies of these tasks are detailed in Table III. Additionally, seven types of robots with varying capabilities, as outlined in Table IV, are considered alongside these construction-related tasks. For instance, robots equipped with a cargo container can assist with material transport and element delivery tasks. Robots with an Indoor Air Quality (IAQ) sensor can help with site inspections. When multiple robots are eligible to perform the same task, robot assignment is determined through the optimization in Sec. III-B, which minimizes the overall task completion time and the number of assigned robots.

C. LLM Performance Analysis

To assess the ability of LLMs to interpret constraints with varying levels of complexity, we collaborated with four construction domain experts to create a test set of 500 descriptive narrative samples (see Supplementary Material: Dataset). Each sample includes one or more types of constraint and parameter changes as defined in Table II. The dataset is evenly divided into five groups based on the number of changes described in each sample. For example, the first group contains 100 samples, each involving a single change. The second group includes 100 samples with two randomly selected changes, and so on, up to the fifth group, which contains samples with five randomly selected changes.

TABLE V
AVERAGE PERFORMANCE ACROSS ALL DIFFICULTY LEVELS

Model	Constraint Acc.	Parameter Acc.	Joint Acc.
claude_haiku	96.2%	74.9%	45.4%
claude_sonnet	96.5%	77.9%	51.4%
gpt4o_mini	96.2%	79.3%	53.8%
gpt4o	99.5%	98.2%	93.4%
gpt4.1_mini	98.9%	96.1%	88.8%
gpt4.1	99.8%	99.2%	97.6%

We then define three metrics to comprehensively evaluate the LLMs' performance, including (1) Constraint Accuracy; (2) Parameter Accuracy; and (3) Joint Accuracy. Constraint Accuracy measures how accurately the LLM identifies the categories of changes, as defined and labeled from 1 to 5 in Table II. Parameter Accuracy evaluates the accuracy of specific extracted values or attributes given each constraint, such as task duration or robot change. Finally, the Joint Accuracy assesses the proportion of cases in which the LLM provides entirely accurate responses across all evaluated constraints and parameters.

To ensure accuracy, the LLM outputs follow a strict JSON-formatted schema defining allowable constraint types and parameter ranges. The prompting process embeds construction-domain knowledge as in-context learning examples, which allows the model to recognize equivalent terminology that may vary among practitioners (e.g., "duct installation" vs. "HVAC fitting") and still map them to the correct standardized task entities. This contextual embedding helps the model interpret site-specific language and measurement units more accurately. With few-shot and chain-of-thought prompting, this approach guides step-by-step reasoning to improve precision and robustness.

The evaluation of the selected LLMs' performance across five task difficulty levels is presented in Fig. 6 and Table V. The results reveal that Constraint Accuracy remains high (close to 100%) across all LLMs, demonstrating their robust ability to interpret and classify task relationships accurately, even as task complexity increases.

The results show that GPT-4.1, GPT-4.1-mini, and GPT-4o outperform Anthropic's Claude models and GPT-4o-mini across five complexity levels. In particular, GPT-4.1 and GPT-4o maintain consistently high accuracy scores (typically above 95%) across difficulty levels. In contrast, Claude models achieve moderate Parameter Accuracy (generally between 70% and 80%). This pattern suggests that Claude and GPT-4o-mini models encounter difficulties extracting precise details, such as task durations and robot types, from complex textual descriptions.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

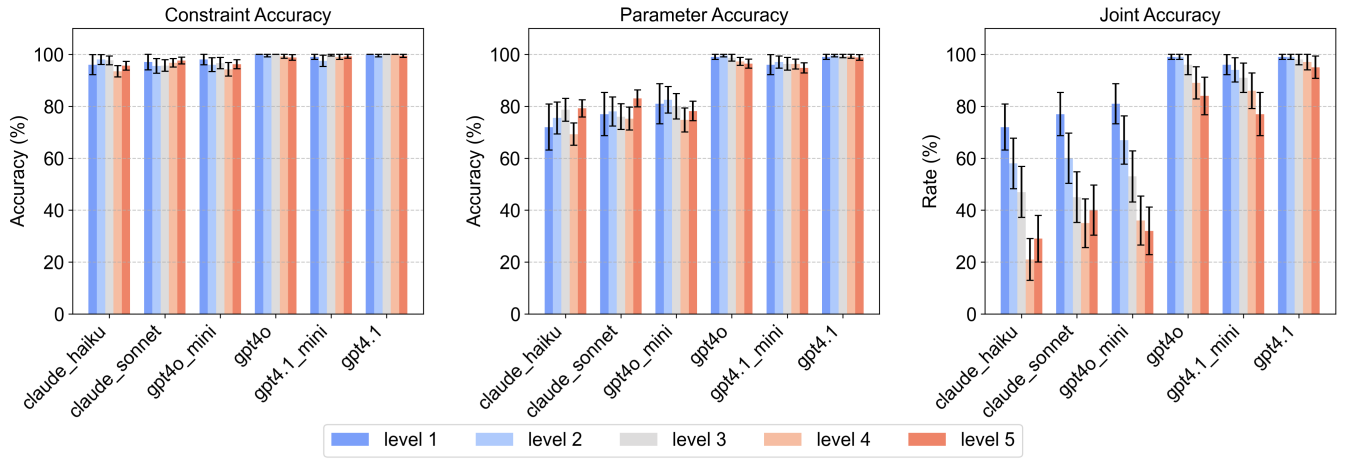


Fig. 6. Performance evaluation of selected LLMs across five complexity levels.

In terms of the Joint Accuracy metric, which provides a more integrated assessment, GPT-4.1 achieves the highest Joint Accuracy across all difficulty levels (consistently above 95%), underscoring its comprehensive understanding and robust performance in combined reasoning tasks. GPT-4o and GPT-4.1-mini also show commendable Joint Accuracy at lower complexity but exhibit noticeable performance drops at higher difficulty. The other three selected LLMs, Claude-Haiku, Claude-Sonnet, and GPT-4o-mini models display substantial reductions in Joint Accuracy, especially at higher complexity levels, often falling below 40%.

Overall, the evaluation identifies OpenAI's GPT-4.1 models as the top performers across all assessed metrics, consistently achieving accuracy higher than 98% with narrower confidence intervals across all five complexity levels. This indicates that GPT-4.1 offers superior precision and reliability in handling intricate construction scheduling analyses compared to other evaluated models. While GPT-4o models also deliver strong performance, they exhibit noticeable limitations as task complexity increases. These findings are promising for construction project scheduling as they illustrate the capability of advanced LLMs, particularly GPT-4.1, to reliably automate and accurately manage complex scheduling tasks.

D. Optimization Results

To evaluate the performance and efficiency of formulated task allocation algorithms in construction tasks, we randomly vary the number of available robots for each type according to the number ranges specified in Table IV. In addition, we vary the number of tasks as outlined in Table III by generating either one or two sets from the groups $\{T_1, T_6, T_7, T_{12}, T_{13}\}$, $\{T_2, T_3, T_8, T_9\}$, and $\{T_4, T_5, T_{10}, T_{11}\}$. Tasks within the same group are selected together to maintain the task dependencies. Overall, 1024 test scenarios with different combinations of robots and tasks were created to evaluate the proposed framework (row 1 in TABLE VI).

To simulate scenarios with additional time window constraints (row 2 in TABLE VI), we randomly select between one and three tasks and impose a condition that these tasks must start after a randomly chosen period of 2 to 4 hours, reflecting scheduling conflicts. For scenarios with task conflict constraints (row 3 in TABLE VI), we enforce that tasks in the set $\{T_6, T_7, T_8, T_9, T_{12}, T_{13}\}$ cannot be performed concurrently, as they require collaboration with a single available worker for these tasks. Similarly, we generated 1024 test scenarios incorporating various combinations of robots and tasks with either the time window or task conflict constraints. To evaluate the replanning optimization (row 4 in TABLE VI), the plan created without time constraints is used as the original plan, and additional time window constraints are introduced during replanning. The updated plan should leverage the original plan while satisfying the time constraints and accounting for the penalty associated with deviations from the original plan.

TABLE VI
EVALUATION OF THE TASK ALLOCATION ALGORITHM

	Tasks	Robots	Max # of Vars	Max # of Cons.	Max Time (sec)	Avg Time (sec)
Original	14-27	7-15	345	377	3.40	0.13
Time window	14-27	7-15	345	379	1.26	0.05
Task conflicts	14-27	7-15	345	378	23.76	0.48
Replanning	14-27	7-15	372	406	0.08	0.02

**The table displays only the maximum and average values across task scenarios. "Vars." and "Cons." denote variables and constraints, respectively.*

The IP-based task allocation is applied to the generated test scenarios. The CP-SAT solver provided by Google OR-Tools is used to solve the integer program. It employs Lazy Clause Generation (LCG) to prune the search space. The solver is run with default parameters and no time limit is imposed: the algorithm terminates only when it finds the optimal solution or exhausts the search space to prove infeasibility. Table VI summarizes the number of tasks, robots, decision variables, and constraints in each scenario, along with the corresponding computational times to find the optimal solution. According to the table, on average, the optimal solutions are found within one second, demonstrating the efficiency of our task allocation formulation. The initial plan can be generated offline, where computation times on the order of seconds remain fully practical. Under real-time construction conditions, replanning

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

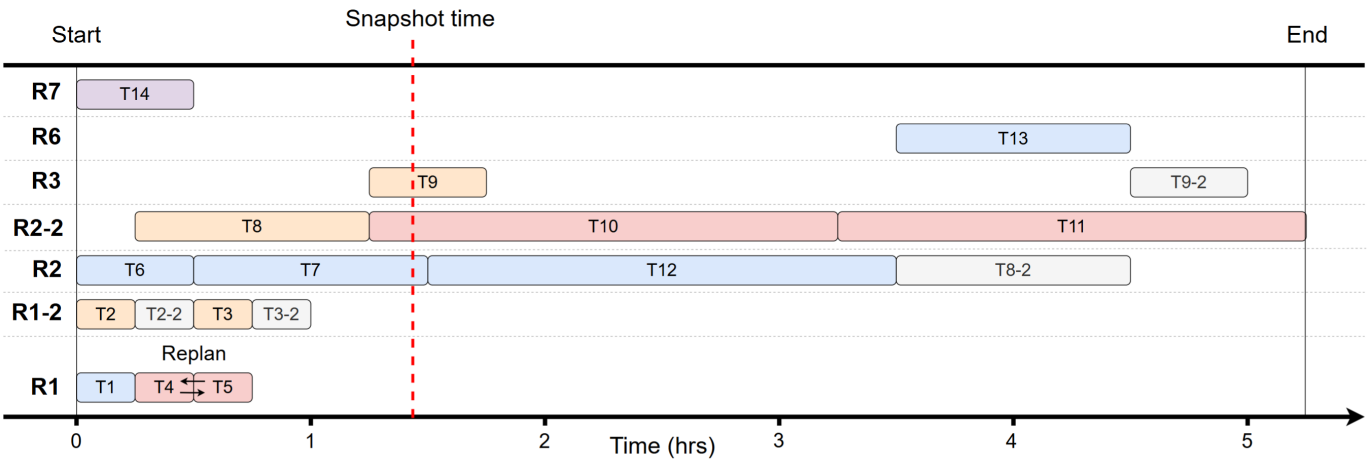


Fig. 7. Example of the optimized task allocation and schedule.

TABLE VII
TASK ALLOCATION USING THE GENETIC ALGORITHM

	Tasks	Robots	Success Rate (%)	Avg Gap (%)	Max Time (sec)	Avg Time (sec)
Original	14-27	7-15	100	6	1.66	1.27
Time window	14-27	7-15	100	14	1.73	1.32
Task conflicts	14-27	7-15	63	10	1.90	1.37
Replanning	14-27	7-15	100	53	2.04	1.45

completes in at most 0.08 seconds. This indicates that the system supports sub-second real-time updates for a comparatively large construction scenario involving 27 tasks and 15 robots.

To evaluate the effectiveness of the proposed integer-programming task allocation algorithm, we compare it with a genetic-algorithm (GA) approach. To maintain computation times on the order of one second, the GA population size is set to 100 with a maximum of 150 generations. The performance results are summarized in Table VII. Under these settings, the GA finds feasible solutions in 63% of the scenarios involving task conflicts and in 100% of the remaining scenarios, whereas the proposed method is guaranteed to return optimal solutions. For the cases in which the GA does find feasible solutions, its average optimality gap relative to the proposed method is consistently positive, as shown in Table VII. Overall, the GA-based task allocation yields a lower success rate, inferior solution quality, and longer computation times compared to the proposed approach.

Fig. 7 presents an example of task allocation results. Following these results, we ran the simulation, and the scene at the snapshot time was captured as shown in Fig. 4 and Fig. 5. In this scenario, the robot fleet includes two robots of type R1, two of type R2, and one each of types R3, R6, and R7. These robots are assigned to complete one set of wall installation and painting tasks, one set of electrical wiring tasks, one set of HVAC duct installation tasks, two sets of window installation tasks, and a site inspection. The optimal completion time for the schedule is calculated to be 5.25 hours.

In addition to the initial plan, a replanned scenario is also illustrated. We present a scenario in which, at 0.2 hours into the schedule, the workers realize that the structural materials required for the HVAC duct installation are unavailable due to delivery delays and will not arrive until 0.5 hours. This disruption necessitates a reallocation of tasks. Applying the replanning framework described in Section III-E, the system minimally adjusts the initial plan to accommodate the disturbance. Considering the penalties for deviating from the original schedule, the updated plan reassigns R1 to execute task T5 before T4, reversing the original task order (as shown in Fig. 7).

V. DISCUSSION

The findings of this study highlight the potential of integrating LLMs, optimization algorithms, and digital twins to address challenges in multi-robot coordination on dynamic construction sites. The results have demonstrated the effectiveness of the adaptive system architecture. By separating the reasoning capabilities of the LLM from the task optimization algorithm itself, the framework preserves flexibility while maintaining computational rigor. The results confirm that the optimization model remains interpretable and robust while benefiting from the LLM's ability to interpret unstructured input and dynamically generate task-specific modifications. Although the accuracy of LLMs in interpreting construction-relevant narratives varies across models, the results underscore their practical viability. In particular, advanced models such as GPT-4.1 consistently extracted constraints and parameters with high accuracy, even as task complexity increased. This suggests strong potential for LLMs to serve as intelligent intermediaries between human planners and autonomous systems, especially in time-sensitive, information-rich environments. Importantly, because the framework is not dependent on a specific LLM architecture, continued advances in LLM capabilities are expected to further enhance its performance and adaptability over time. At the system level, the integration of a digital twin enabled a closed-loop information flow between the physical construction site and its virtual representation. Real-time updates of task progress, robot states, and environmental changes are synchronized through the digital twin,

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

which allows for continuous refinement of task allocations and site-level decision-making. This capability is especially valuable in dynamic construction settings, where conditions shift frequently and adaptability is paramount. The simulation results have supported the practicability of the proposed framework.

Through the digital twin interface, users can monitor task progress, visualize robot operations, and intervene when unexpected events occur (e.g., weather disruption or priority change). Human input is given in natural language and processed by the LLM interpreter, which translates these instructions into formalized constraint updates for the IP optimizer. The optimizer then recalculates the task allocation plan automatically, ensuring that user decisions are consistently propagated through the system without manual reprogramming.

Beyond the context of construction, the proposed framework exhibits strong potential for scalability and cross-domain applications. The generalizable principles of optimization, real-time feedback, and natural language reasoning, make it highly adaptable to other sectors such as manufacturing, logistics, warehousing, and industrial assembly. In these domains, similar to construction, tasks often follow strict sequences, depend on heterogeneous agent capabilities, and may be subject to unexpected disruptions. Additionally, the digital twin integration offers significant benefits for sectors that rely heavily on spatial coordination and progress tracking, such as infrastructure development and maintenance.

Despite some promising outcomes, there are several limitations in this study. For example, the current simulation setup simplifies certain aspects of real-world construction scenarios. In particular, the collaboration between robots and human workers is not explicitly modeled, resulting in idealized task transitions that may differ slightly from real-world conditions. While the current study evaluates a representative construction scenario, extending the framework to additional cases remains an important direction for future work. These scenarios can be incorporated by adjusting BIM task graphs, robot configurations, and constraint definitions within the digital twin, without modifying the core decision-making architecture. Additionally, although LLMs demonstrate strong performance under structured prompting, their role in this framework is currently limited. Future work will extend the LLM toward proactive conflict detection and pre-optimization reasoning. This extension will allow the framework to identify potential inconsistencies before solver execution while preserving transparency and mathematical rigor.

VI. Conclusion

This research introduced an integrated framework for adaptive multi-robot task allocation in dynamic construction environments, combining digital twin technology, IP optimization, and LLM-driven narrative interpretation. The proposed system addresses the challenges posed by real-time uncertainties, enabling efficient and flexible task rescheduling without manual intervention. By decoupling the reasoning and optimization processes, the framework maintains algorithmic transparency and scalability, offering a structured and scalable solution adaptable to evolving site conditions. Empirical evaluations demonstrated the computational robustness of the optimization model. They also showed that LLMs can effectively extract constraint modifications from unstructured natural language descriptions with high accuracy. The establishment of a closed-loop feedback mechanism between the physical construction site and its digital twin representation further enhances situational awareness and supports continuous system adaptation. Overall, the proposed framework demonstrates the potential of integrating LLMs, optimization, and digital twins to support intelligent multi-robot operations in construction and other industrial applications.

References

[1] H. Begić, M. Galić, and Z. Dolaček-Alduk, "Digitalization and automation in construction project's life-cycle: A review," *J. Inf. Technol. Construct.*, vol. 27, pp. 441–460, Apr. 2022.

[2] Q. Chen, B. García de Soto, and B. T. Adey, "Construction automation: Research areas, industry concerns and suggestions for advancement," *Autom. Construct.*, vol. 94, pp. 22–38, Oct. 2018.

[3] A. Klarin and Q. Xiao, "Automation in architecture, engineering and construction: A scientometric analysis and implications for management," *Eng., Construct. Architectural Manage.*, vol. 31, no. 8, pp. 3308–3334, Jul. 2024.

[4] K. S. Saidi, T. Bock, and C. Georgoulas, "Robotics in construction," in *Springer Handbook of Robotics*. Cham, Switzerland: Springer, 2016, pp. 1493–1520.

[5] S. Liu, X. V. Wang, and L. Wang, "Digital twin-enabled advance execution for human–robot collaborative assembly," *CIRP Ann.*, vol. 71, no. 1, pp. 25–28, 2022.

[6] X. Wang, S. Wang, C. C. Menassa, V. R. Kamat, and W. McGee, "Automatic high-level motion sequencing methods for enabling multi-tasking construction robots," *Autom. Construct.*, vol. 155, Nov. 2023, Art. no. 105071.

[7] B. Fu et al., "Digital twin-based smart manufacturing: Dynamic line reconfiguration for disturbance handling," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 14892–14905, 2025.

[8] B. M. Tehrani, S. BuHamdan, and A. Alwisy, "Robotics in assembly-based industrialized construction: A narrative review and a look forward," *Int. J. Intell. Robot. Appl.*, vol. 7, no. 3, pp. 556–574, Sep. 2023.

[9] J. Xu, M. Dixit, and X. Wang, "Adaptive visual perception for robotic construction process: A multi-robot coordination framework," 2024, *arXiv:2412.11275*.

[10] M. Gharbia, A. Chang-Richards, Y. Lu, R. Y. Zhong, and H. Li, "Robotic technologies for on-site building construction: A systematic review," *J. Building Eng.*, vol. 32, Nov. 2020, Art. no. 101584.

[11] A. Attalla, O. Attalla, A. Moussa, D. Shafique, S. B. Raean, and T. Hegazy, "Construction robotics: Review of intelligent features," *Int. J. Intell. Robot. Appl.*, vol. 7, no. 3, pp. 535–555, Sep. 2023.

[12] B. Xiao, C. Chen, and X. Yin, "Recent advancements of robotics in construction," *Autom. Construct.*, vol. 144, Dec. 2022, Art. no. 104591.

[13] B. Fu, W. Smith, D. M. Rizzo, M. Castanier, M. Ghaffari, and K. Barton, "Robust task scheduling for heterogeneous robot teams under capability uncertainty," *IEEE Trans. Robot.*, vol. 39, no. 2, pp. 1087–1105, Apr. 2023.

[14] O. Valero, J. Antich, A. Tauler-Rosselló, J. Guerrero, J.-J. Miñana, and A. Ortiz, "Multi-robot task allocation methods: A fuzzy optimization approach," *Inf. Sci.*, vol. 648, Nov. 2023, Art. no. 119508.

[15] T. Lei, P. Chintam, C. Luo, L. Liu, and G. E. Jan, "A convex optimization approach to multi-robot task allocation and path planning," *Sensors*, vol. 23, no. 11, p. 5103, May 2023.

[16] B. Abdzadeh, S. Noori, and S. F. Ghannadpour, "Simultaneous scheduling of multiple construction projects considering supplier selection and material transportation routing," *Autom. Construct.*, vol. 140, Aug. 2022, Art. no. 104336.

[17] L. A. Wolsey, *Integer Programming*. Hoboken, NJ, USA: Wiley, 2020.

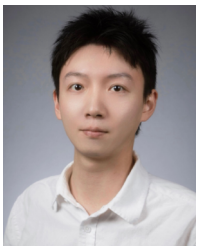
[18] F. A. Omara and M. M. Arafa, "Genetic algorithms for task scheduling problem," *J. Parallel Distrib. Comput.*, vol. 70, no. 1, pp. 13–22, Jan. 2010.

[19] K. Wang and A. Jacquillat, "A stochastic integer programming approach to air traffic scheduling and operations," *Oper. Res.*, vol. 68, no. 5, pp. 1375–1402, Sep. 2020.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.

[20] C. Perreault-Lafleur, M. Carvalho, and G. Desaulniers, "A stochastic integer programming approach to reserve staff scheduling with preferences," *Int. Trans. Oper. Res.*, vol. 32, no. 1, pp. 289–313, Jan. 2025.
[21] L. R. Abreu, J. O. Cunha, B. A. Prata, and J. M. Framinan, "A genetic algorithm for scheduling open shops with sequence-dependent setup times," *Comput. Oper. Res.*, vol. 113, Jan. 2020, Art. no. 104793.
[22] J. Deng, "Research on the risk early warning of construction engineering under the coupling disaster of typhoons and rainstorms in coastal areas based on BP neural network," *J. Coastal Res.*, vol. 105, no. sp1, pp. 151–154, Dec. 2020.
[23] S. J. Schuldt, M. R. Nicholson, Y. A. Adams, and J. D. Delorit, "Weather-related construction delays in a changing climate: A systematic state-of-the-art review," *Sustainability*, vol. 13, no. 5, p. 2861, Mar. 2021.
[24] D. Yu, Q. Lv, G. Srivastava, C.-H. Chen, and J. C.-W. Lin, "Multiobjective evolutionary model of the construction industry based on network planning," *IEEE Trans. Ind. Informat.*, vol. 19, no. 2, pp. 2173–2182, Jul. 2022.
[25] J. Lee, M. Park, H.-S. Lee, T. Kim, S. Kim, and H. Hyun, "Workflow dependency approach for modular building construction manufacturing process using dependency structure matrix (DSM)," *KSCE J. Civil Eng.*, vol. 21, no. 5, pp. 1525–1535, Jul. 2017.
[26] T. Jin, Z. Cai, B. Li, C. Zheng, G. Jiang, and J. Cheng, "Improving resource utilization by timely fine-grained scheduling," in *Proc. 15th Eur. Conf. Comput. Syst.*, Apr. 2020, pp. 1–16.
[27] H. Naveed et al., "A comprehensive overview of large language models," 2023, *arXiv:2307.06435*.
[28] I. Armeni et al., "Construction robotics and automation [TC spot light]," *IEEE Robot. Autom. Mag.*, vol. 31, no. 4, pp. 186–192, Dec. 2024.
[29] H. Yu, V. R. Kamat, and C. C. Menassa, "Generalizable skill learning for construction robots with crowdsourced natural language instructions, composable skills standardization, and a large language model," *ASCE OPEN, Multidisciplinary J. Civil Eng.*, vol. 3, no. 1, Dec. 2025, Art. no. 04025014.
[30] J. Külz, M. Terzer, M. Magri, A. Giusti, and M. Althoff, "Holistic construction automation with modular robots: From high-level task specification to execution," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 16716–16727, 2025.
[31] M. Deng, C. C. Menassa, and V. R. Kamat, "From BIM to digital twins: A systematic review of the evolution of intelligent building representations in the AEC-FM industry," *J. Inf. Technol. Construct.*, vol. 26, pp. 58–83, Feb. 2021.
[32] X. Wang, C.-J. Liang, C. C. Menassa, and V. R. Kamat, "Interactive and immersive process-level digital twin for collaborative human–robot construction work," *J. Comput. Civil Eng.*, vol. 35, no. 6, Nov. 2021, Art. no. 04021023.
[33] S. Kim, M. Peavy, P.-C. Huang, and K. Kim, "Development of BIM-integrated construction robot task planning and simulation system," *Autom. Construct.*, vol. 127, Jul. 2021, Art. no. 103720.
[34] X. Wang, H. Yu, W. McGee, C. C. Menassa, and V. R. Kamat, "Enabling building information model-driven human–robot collaborative construction workflows with closed-loop digital twins," *Comput. Ind.*, vol. 161, Oct. 2024, Art. no. 104112.
[35] X. Lin, Z. Guo, X. Jin, and H. Guo, "Digital twin-enabled safety monitoring system for seamless worker-robot collaboration in construction," *Autom. Construct.*, vol. 174, Jun. 2025, Art. no. 106147.
[36] H. Chakraa, F. Guérin, E. Leclercq, and D. Lefebvre, "Optimization techniques for multi-robot task allocation problems: Review on the state-of-the-art," *Robot. Auto. Syst.*, vol. 168, Oct. 2023, Art. no. 104492.
[37] X. Bai, A. Fielbaum, M. Kronmüller, L. Knoedler, and J. Alonso-Mora, "Group-based distributed auction algorithms for multi-robot task assignment," *IEEE Trans. Autom. Sci. Eng.*, vol. 20, no. 2, pp. 1292–1303, Apr. 2023.
[38] Á. Calvo and J. Capitán, "Heterogeneous multirobot task allocation for long-endurance missions in dynamic scenarios," *IEEE Trans. Robot.*, vol. 41, pp. 6494–6513, 2025.
[39] H. Guo, Z. Miao, J. Ji, and Q. Pan, "An effective collaboration evolutionary algorithm for multi-robot task allocation and scheduling in a smart farm," *Knowl.-Based Syst.*, vol. 289, Apr. 2024, Art. no. 111474.
[40] Z. Pan and Y. Yu, "Optimizing heterogeneous multi-robot team composition for long-horizon construction tasks: Time- and utilization-guided simulation," *Autom. Construct.*, vol. 165, Sep. 2024, Art. no. 105520.
[41] D. Lee, "Resource-based task allocation for multi-robot systems," *Robot. Auto. Syst.*, vol. 103, pp. 151–161, May 2018.
[42] J. G. Martin, F. J. Muros, J. M. Maestre, and E. F. Camacho, "Multi-robot task allocation clustering based on game theory," *Robot. Auto. Syst.*, vol. 161, Mar. 2023, Art. no. 104314.
[43] F. Yan and K. Di, "Solving the multi-robot task allocation with functional tasks based on a hyper-heuristic algorithm," *Appl. Soft Comput.*, vol. 146, Oct. 2023, Art. no. 110628.
[44] X. Bai, Y. Ye, B. Zhang, and S. S. Ge, "Efficient package delivery task assignment for truck and high capacity drone," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 13422–13435, Nov. 2023.
[45] X. Bai et al., "Efficient hybrid multi-population genetic algorithm for multi-UAV task assignment in consumer electronics applications," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 2395–2406, May 2025.
[46] M. L. Betalo, S. Leng, A. M. Seid, H. N. Abishu, A. Erbad, and X. Bai, "Dynamic charging and path planning for UAV-powered rechargeable WSNs using multi-agent deep reinforcement learning," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 15610–15626, 2025.
[47] M. Leranso Betalo, I. Ullah, F. Berhanu Tesema, Z. Wu, J. Li, and X. Bai, "Generative AI-driven multiagent DRL for task allocation in UAV-assisted EMPD within 6G-enabled SAGIN networks," *IEEE Internet Things J.*, vol. 12, no. 17, pp. 35890–35907, Sep. 2025.
[48] Y. Wang, H. Luo, and W. Fang, "An integrated approach for automatic safety inspection in construction: Domain knowledge with multimodal large language model," *Adv. Eng. Informat.*, vol. 65, May 2025, Art. no. 103246.
[49] Z. Qian and C. Shi, "Large language model-empowered paradigm for automated geotechnical site planning and geological characterization," *Autom. Construct.*, vol. 173, May 2025, Art. no. 106103.
[50] H. Pu, X. Yang, J. Li, and R. Guo, "AutoRepo: A general framework for multimodal LLM-based automated construction reporting," *Expert Syst. Appl.*, vol. 255, Dec. 2024, Art. no. 124601.
[51] G. Chen, A. Alsharef, A. Ovid, A. Albert, and E. Jaselskis, "Meet2Mitigate: An LLM-powered framework for real-time issue identification and mitigation from construction meeting discourse," *Adv. Eng. Informat.*, vol. 64, Mar. 2025, Art. no. 103068.
[52] K. Forth and A. Borrmann, "Semantic enrichment for BIM-based building energy performance simulations using semantic textual similarity and fine-tuning multilingual LLM," *J. Building Eng.*, vol. 95, Oct. 2024, Art. no. 110312.
[53] S. A. Prieto, E. T. Mengiste, and B. García de Soto, "Investigating the use of ChatGPT for the scheduling of construction projects," *Buildings*, vol. 13, no. 4, p. 857, Mar. 2023.
[54] K. Kim, M. Ivashchenko, P. Ghimire, and P.-C. Huang, "Context-aware and adaptive task planning for autonomous construction robots through llm-robot communication," *Available SSRN J.*, 2024.
[55] J. White et al., "A prompt pattern catalog to enhance prompt engineering with chatgpt," 2023, *arXiv:2302.11382*.
[56] S. Atreja, J. Ashkinaze, L. Li, J. Mendelsohn, and L. Hemphill, "Prompt design matters for computational social science tasks but in unpredictable ways," 2024, *arXiv:2406.11980*.
[57] M. Bosma et al., "Chain-of-thought prompting elicits reasoning in large language models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 24824–24837.
[58] T. B. Brown et al., "Language models are few-shot learners," in *Proc. NIPS*, 2020, pp. 1877–1901.
[59] S. Hegselmann, A. Buendia, H. Lang, M. Agrawal, X. Jiang, and D. Sontag, "TabLLM: Few-shot classification of tabular data with large language models," in *Proc. Int. Conf. Artif. Intell. Statist. (PMLR)*, 2023, pp. 5549–5581.
[60] D. Hendrycks et al., "Measuring massive multitask language understanding," 2020, *arXiv:2009.03300*.
[61] L. Fan, W. Hua, L. Li, H. Ling, and Y. Zhang, "NPHardEval: Dynamic benchmark on reasoning ability of large language models via complexity classes," 2023, *arXiv:2312.14890*.
[62] Y. Sonoda et al., "Diagnostic performances of GPT-4o, Claude 3 opus, and Gemini 1.5 pro in 'diagnosis please' cases," *Jpn. J. Radiol.*, vol. 42, no. 11, pp. 1231–1235, Nov. 2024.
[63] M. O. Sanni-Anibire, R. Mohamad Zin, and S. O. Olatunji, "Causes of delay in the global construction industry: A meta analytical review," *Int. J. Construct. Manage.*, vol. 22, no. 8, pp. 1395–1407, Jun. 2022.
[64] R. S. Lopes, C. A. V. Cavalcante, and M. H. Alencar, "Delay-time inspection model with dimensioning maintenance teams: A study of a company leasing construction equipment," *Comput. Ind. Eng.*, vol. 88, pp. 341–349, Oct. 2015.
[65] S. Durdyev and M. R. Hosseini, "Causes of delays on construction projects: A comprehensive list," *Int. J. Manag. Projects Bus.*, vol. 13, no. 1, pp. 20–46, Jan. 2020.
[66] *Siemens/Ros-Sharp*. Accessed: Jun. 10, 2025. [Online]. Available: https://github.com/siemens/ros-sharp/wiki
[67] R. Rust et al.. (2018). *COMPAS FAB: Robotic Fabrication Package for the COMPAS Framework*. [Online]. Available: https://github.com/compas-dev/compas_fab/

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.



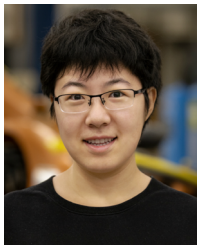
Min Deng (Member, IEEE) received the B.E. degree (Hons.) in architectural environment engineering from the University of Nottingham Ningbo, China, in 2015, the M.S. degree in intelligent building technology and the M.Phil. degree in civil engineering from The Hong Kong University of Science and Technology in 2016 and 2018, respectively, and the M.S. degree in robotics and the Ph.D. degree in civil engineering from the University of Michigan, Ann Arbor, in 2023. He is currently an Assistant Professor with the Department of Civil and Environmental Engineering, University of Tennessee, Knoxville. His research interests include embodied AI, human–robot collaboration, coordinated multi-robot systems, digital twins, and robot learning for adaptive autonomy in complex and unstructured environments.



Bo Fu received the B.S. degree in vehicle engineering from Tongji University, China, in 2017, the M.S. degree in mechanical engineering from Carnegie Mellon University in 2019, and the Ph.D. degree in robotics from the University of Michigan in 2024. He is currently an Applied Scientist with Amazon Robotics. His research interests include robust and resilient decision-making for heterogeneous multi-agent systems under uncertainty using optimization and learning-based approaches. The application areas include large-scale multi-robot collaboration, multi-agent path finding, warehouse management, human–robot interaction, robotic construction, and smart manufacturing systems.



Lingyao Li received the B.S. and M.S.E. degrees in ocean engineering from Harbin Engineering University in 2012 and 2014, respectively, and the Ph.D. degree in civil and environmental engineering from the University of Maryland in 2021. Subsequently, he completed post-doctoral training with the University of Maryland in 2022 and the University of Michigan in 2024. He is currently an Assistant Professor with the School of Information, University of South Florida. His research interests center on AI agents and large language models, with application areas, including health systems and urban systems.



Xi Wang (Member, IEEE) received the B.S. degree in construction management from Dongbei University of Finance and Economics, China, in 2017, the M.S.E. degree in construction engineering and management from the University of Michigan, Ann Arbor, in 2018, and the M.S. degree in robotics and the Ph.D. degree in civil engineering from the University of Michigan, Ann Arbor, in 2022. She is currently an Assistant Professor with the Department of Construction Science, Texas A&M University. Her research interests include context-aware robotic construction, human–robot interaction, and multi-robot coordination. She is the Co-Chair of IEEE RAS Technical Committee on Construction Robotics.

Authorized licensed use limited to: CITY UNIV OF HONG KONG. Downloaded on August 12,2026 at 15:34:18 UTC from IEEE Xplore. Restrictions apply.